

Week 1

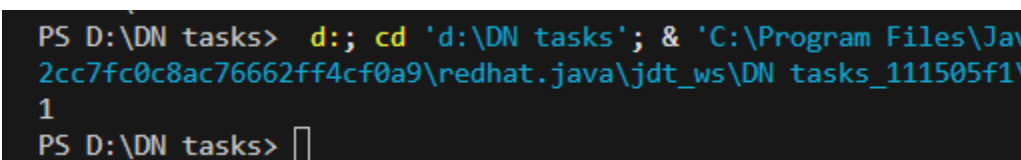
Exercise 3: E-commerce Platform Search Function

Code:

A screenshot of an IDE window with multiple tabs. The active tab is 'Ecomm.java 2, U'. The code is in Java and defines a 'Product' class and an 'Ecomm' class. The 'Product' class has attributes 'id' and 'name', and a constructor. The 'Ecomm' class has a static method 'linearSearch' that iterates through an array of 'Product' objects to find one with a matching name (case-insensitive). It also has a 'main' method that creates an array of three products (Mobile, Laptop, Watch) and calls 'linearSearch' with 'Laptop' as the key, printing the result.

```
J Main.java J Factory.java 2 J Ecomm.java 2, U X J Forecast.java U
J Ecomm.java > Java > Ecomm > linearSearch(Product[] arr, String key)
1 class Product {
2     int id;
3     String name;
4
5     Product(int id, String name) {
6         this.id = id;
7         this.name = name;
8     }
9 }
10
11 public class Ecomm {
12
13     public static int linearSearch(Product arr[], String key) {
14
15         for(int i=0;i<arr.length;i++) {
16             if(arr[i].name.equalsIgnoreCase(key))
17                 return i;
18         }
19
20         return -1;
21     }
22
23     Run main | Debug main | Run | Debug
24     public static void main(String[] args) {
25
26         Product products[] = {
27             new Product(id: 1,name: "Mobile"),
28             new Product(id: 2,name: "Laptop"),
29             new Product(id: 3,name: "Watch")
30         };
31
32         int result = linearSearch(products,key: "Laptop");
33
34         System.out.println(result);
35     }
36 }
```

Output:

A screenshot of a terminal window. The prompt is 'PS D:\DN tasks>'. The user enters a command to run a Java program. The output of the program is '1', which is printed on the line following the command. The prompt then shows a cursor waiting for the next input.

```
PS D:\DN tasks> d:;; cd 'd:\DN tasks'; & 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' -cp 'C:\Program Files\Java\jdk-11.0.2\bin\java.exe' 2cc7fc0c8ac76662ff4cf0a9\redhat.java\jdt_ws\DN_tasks_111505f1\1
1
PS D:\DN tasks> 
```